

G: Guard Deployment

Source file name: `guard.c`, `guard.cpp`, `guard.java`, or `guard.py`

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The Nlogonian Army military base is located in a remote and strategic area. The base is composed of a collection of rectangular buildings with flat roofs. The buildings are arranged in a way that creates an irregularly shaped base. Each building is defined by its base (a rectangle in the 2D plane) and its height (all in meters).

To increase the security of the base, the commander has decided to build a convex polygonal perimeter fence with *guard towers* at each vertex and surveillance cameras on their tops. The commander wants to minimize the length of the fence to save resources.

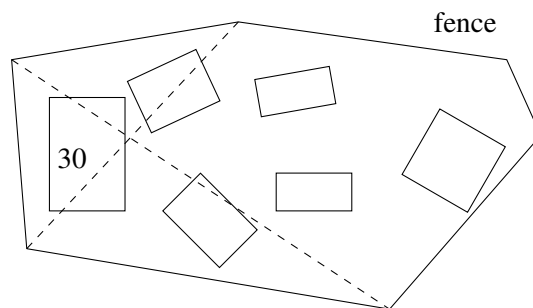
Additionally, the guard towers must have a line of sight to enable swift communication between the guards in an emergency. A line of sight between two guard towers is ensured if no part of the line intersects or touches the surface or edge of any building, including its roof. Since unexpected objects may be found on the roof of some buildings, if a line of sight passes above a building, it must do so at least one meter higher than the building's roof.

To facilitate communication, surveillance and manufacture, the commander also wants that all the designed guard towers has the same height. Furthermore, he wants this height to be minimal to avoid making the guard towers too visible and to save resources.

The commander has asked his team of engineers to determine the optimal length of the perimeter fence and the minimum height of the guard towers that guarantees a line of sight between them.

Can you help the team of engineers solving this problem?

The following figure shows the layout of a military base with six rectangular buildings. The building on the left is 30 meters high, and assume all the others are 10 meters high. Also shown is a polygonal perimeter fence (not necessarily optimal) surrounding the base. The two dotted lines show two lines of sight, justifying the placement of the cameras at a height of 31 meters.



Input

The first line contains a number K ($1 \leq K \leq 10^4$), the number of test cases.

For each case, the first line contains a natural number N ($1 \leq N \leq 10^3$) representing the number of buildings in the base. Each of the following N lines contains nine integers: $x_1, y_1, x_2, y_2, x_3, y_3, x_4, y_4, h$. The points (x_1, y_1) , (x_2, y_2) , (x_3, y_3) and (x_4, y_4) are the coordinates of the corners of the rectangular building's base ($0 \leq |x_1|, |y_1|, |x_2|, |y_2|, |x_3|, |y_3|, |x_4|, |y_4| \leq 10^6$), and, h ($1 \leq h \leq 10^3$) is the height of the building.

It is guaranteed that the input data correspond to rectangular bases and that no two of them overlap. More exactly, for each pair pairs of buildings, the intersection of their bases measures 0.

The input must be read from standard input.

Output

A single line containing two numbers separated by a single space: the optimal perimeter of the fence (a floating-point number with 6 decimal places) and the minimum height of the guard towers that ensures visibility between all pairs of towers.

The minimum height at which a watchtower must be located is 3 meters.

The output must be written to standard output.

Sample Input	Sample Output
1 5 3 0 5 2 2 5 0 3 10 2 6 5 6 5 8 2 8 5 9 8 8 9 7 8 8 7 30 6 5 8 5 8 6 6 6 31 6 3 8 3 8 4 6 4 20	28.054753 32